Mechanics 1
Pin jointed trusses
Part 1- Method of Joints

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CONTENT

- Trusses recap
- Method of joints

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Examples of truss structures

The Forth Bridge, Scotland

The General Hertzog Bridge over the Orange River, South Africa.

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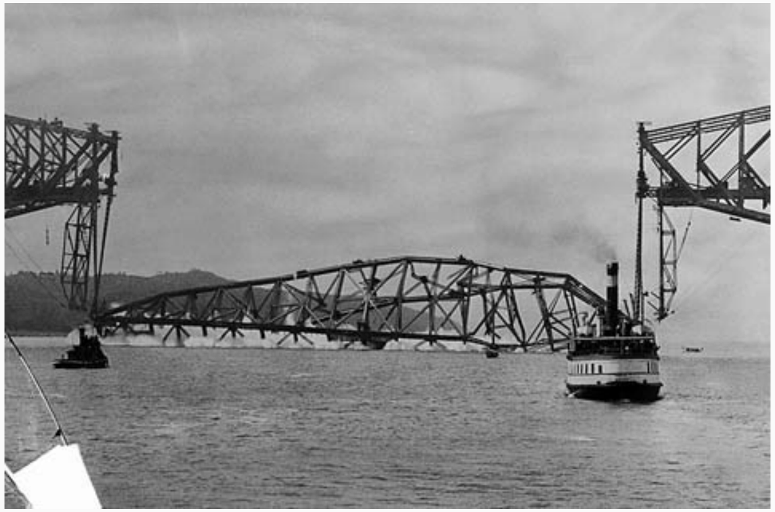
TRUSS MEMBERS CAN ONLY TRANSMIT AXIAL LOADS

Examples of truss structures

August, 2007. Minneapolis I-35W bridge (Minneapolis, USA)
During the evening rush hour, the main spans of the bridge collapsed into the Mississippi River.



September, 1916. Quebec Bridge (Canada). Collapse is due to distortions of key structural members.

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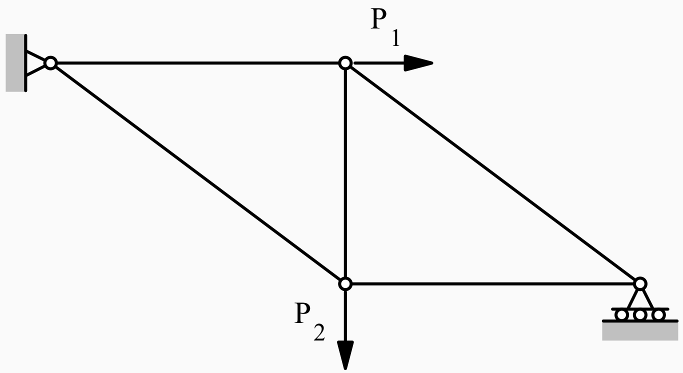
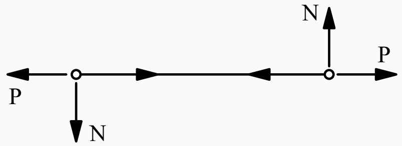
Pin-jointed trusses--recap

A class of structures in which:



(i) Straight bar members are connected together at 'frictionless' pin joints.
(ii) External loads are applied <u>only</u> at the joints

Note that this definition implies that internal loads must consist only of <u>axial</u> forces in the members.



Forces acting on the ends of an individual member must be equal and opposite - and for rotational equilibrium, N = 0. Hence the resultant must be parallel to the axis of the member - and the internal force in the member must be axial.

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Method of joints

Finding all of the external reactions and all of the internal forces can be carried out by the <u>Method of Joints</u>.

1) Using overall equilibrium equations find the external reactions.
   $\Sigma V=0$, $\Sigma H=0$, $\Sigma M=0$ for overall structure.
2) Select a joint at which there are no more than two unknown forces acting on the pin and solve for the unknown forces.
   $\Sigma V=0$, $\Sigma H=0$ for joint
3) Note that the forces acting on the pins at the other ends of the members are equal and opposite to those found in (ii) - and mark in accordingly
4) Repeat from step (ii) until all internal forces have been found.

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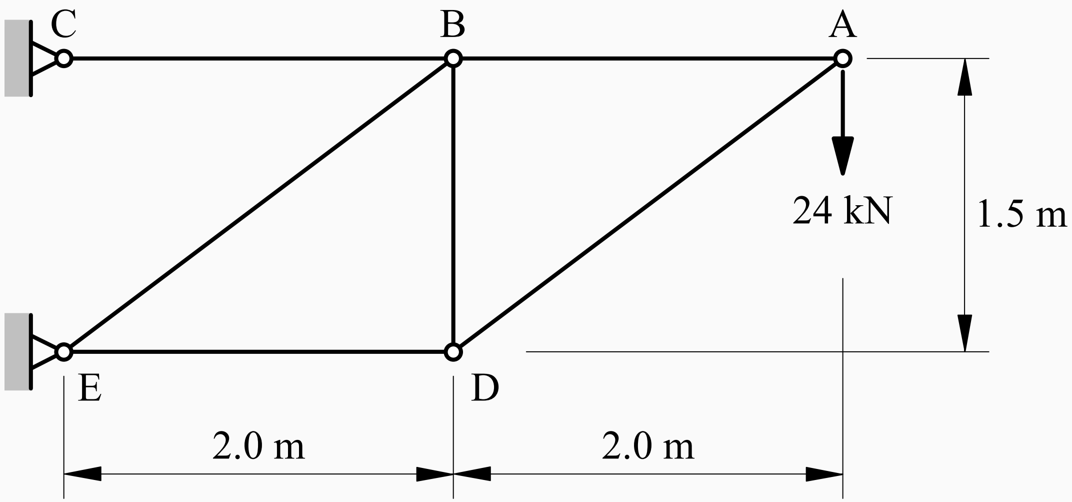
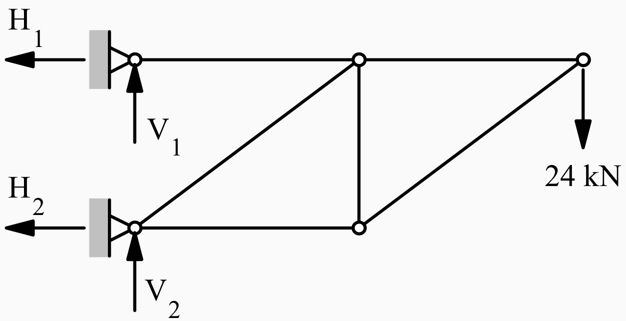
Example 1

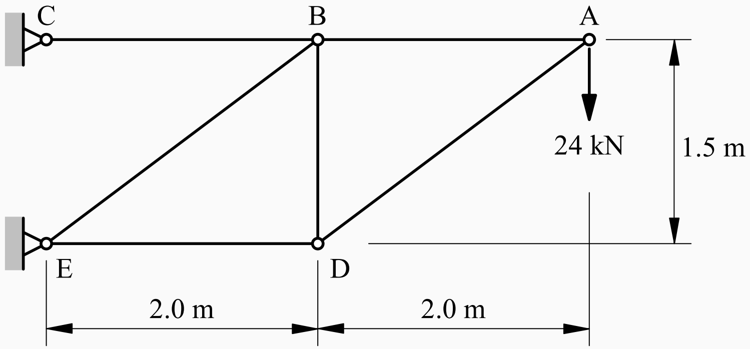
The truss shown in the diagram is pinned to a wall at C and E, and carries a vertical force of 24kN at A. What are the forces on the wall and what are the internal forces in the members?



Example 1--External reactions

Clearly there are four possible reaction components - two at each of the pin supports.





For overall equilibrium :

$\Sigma V = 0$: $V_1 + V_2 - 24 = 0$

$\Sigma H = 0$: $H_1 + H_2 = 0$

$\Sigma M = 0$ $(H_1 \times 1.5) - (24 \times 4) = 0$ $\therefore H_1 = 64$kN

Consider rotational equilibrium of BC: $(V_1 \times 2) = 0$ $\therefore V_1 = 0$

$V_1 = 0$ kN

$V_2 = 24$ kN

$H_1 = 64$ kN

$H_2 = -64$ kN

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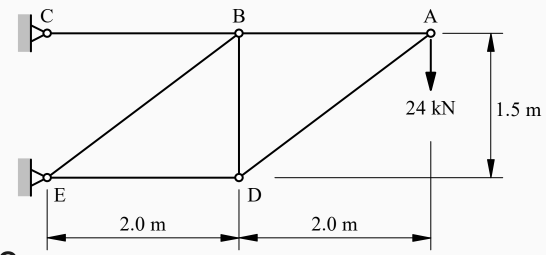
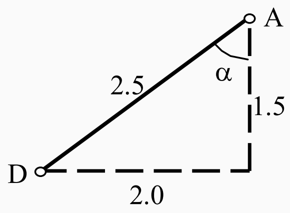
Example 1--Internal forces

Assume that all the members are in tension. Then identify a joint at which there are no more than two unknown forces.



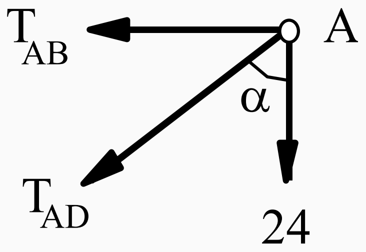
Consider equilibrium of forces acting on the pin at Joint A.

Note that the sine and cosine of the angle at which member AD lies can be read directly from the drawing :



$\sin\alpha = 2.0/2.5 = 0.8$

$\cos\alpha = 1.5/2.5 = 0.6$



$\Sigma V = 0$

$$24 + T_{AD}\cos\alpha = 0$$

$$\therefore\ T_{AD} = -24/0.6 = -40\text{kN}$$

$\Sigma H = 0$:

$$T_{AB} + T_{AD}\sin\alpha = 0$$

$$\therefore\ T_{AB} = 40 \times 0.8 = 32 \text{ kN}$$

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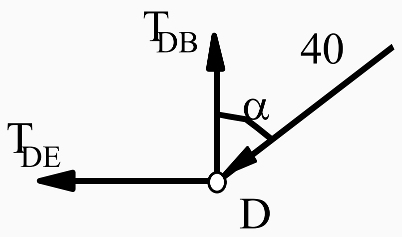
Example 1--Internal forces

Now proceed to Joint D - and consider equilibrium of forces acting on the pin :

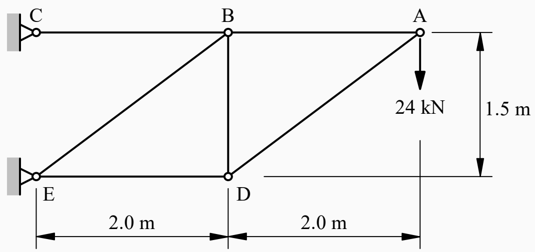


$\Sigma V = 0$: $\quad 40\cos\alpha - T_{DB} = 0$

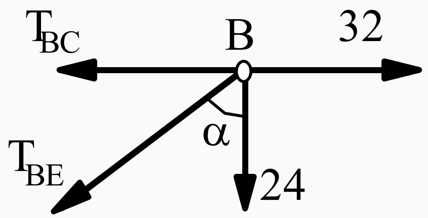
$\therefore T_{DB} = 40 \times 0.6 = 24$ kN

$\Sigma H = 0$: $\quad 40\sin\alpha + T_{DE} = 0$

$\therefore T_{DE} = -40 \times 0.8 = -32$ kN



And finally, at Joint B :



$\Sigma V = 0$: $\quad 24 + T_{BE}\cos\alpha = 0$

$\therefore T_{BE} = -24 / 0.6 = -40$ kN

$\Sigma H = 0$: $\quad T_{BC} + T_{BE}\sin\alpha - 32 = 0$

$\therefore T_{BC} = 32 + 40 \times 0.8 = 64$ kN

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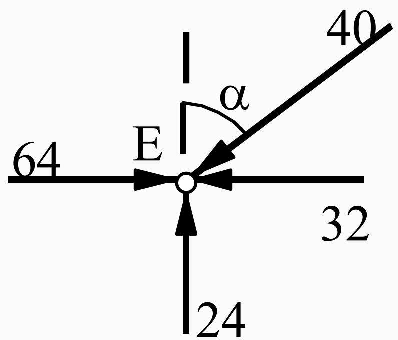
Example 1--Internal forces

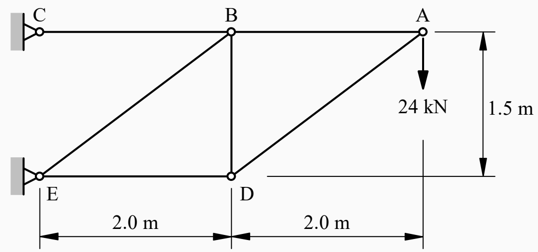
All of the internal forces have been found - but it is a good idea to confirm the calculations by checking equilibrium at the last joint.

Thus, at Joint E :



$\Sigma V = 0;$ $\quad 24 - 40 \times \cos\alpha = 0$ ✓ check

$\Sigma H = 0;$ $\quad 64 - 32 - 40 \times \sin\alpha = 0$ ✓ check



Member	AB	AD	BC	BD	BE	DE
Axial Force (kN)	+32	-40	+64	+24	-40	-32

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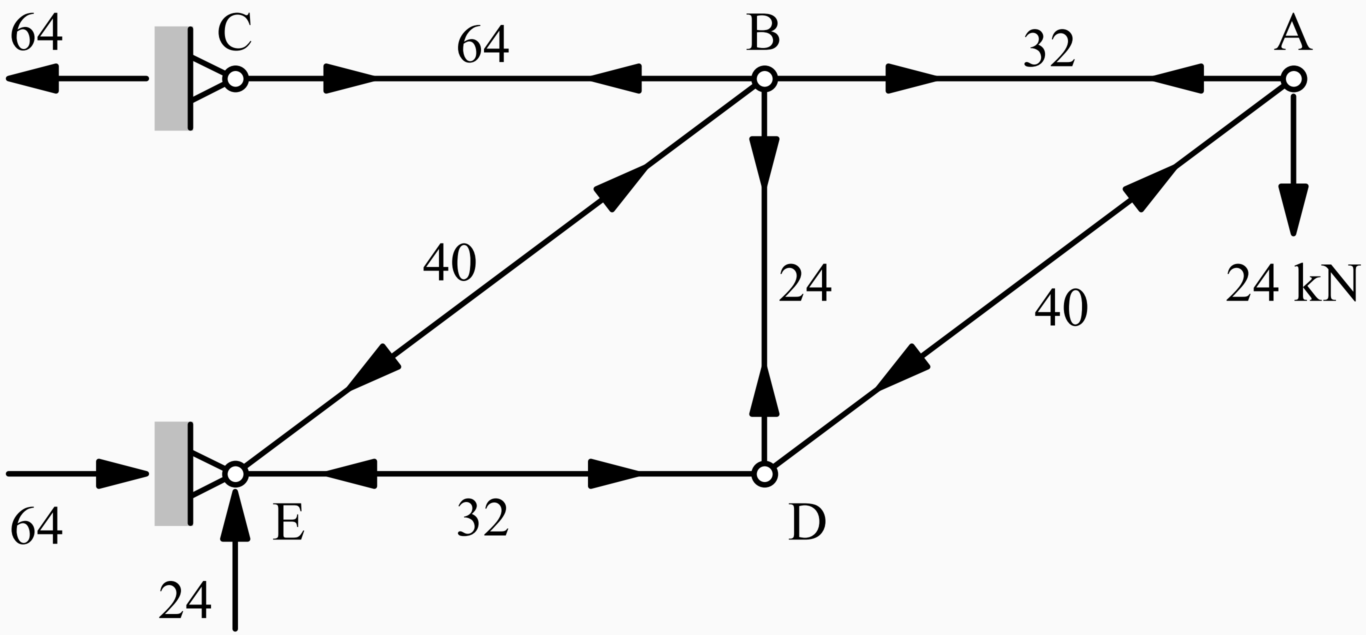
Example 1--All forces



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What have we looked at?

External and internal forces

Pin-jointed trusses

You now should be able to attempt tutorial 2